STATISTICAL MODELLING AND TREND ANALYSIS OF URBAN CRIME PATTERNS USING LARGE SCALE CRIME DATA

Submitted by

Helen Maria Ajay

Register No: 22MIY0057

Non-Parametric Statistics Course Project

Winter Semester 2025-2026

Abstract

Urban crime is one of the most pressing challenges facing modern cities, and understanding the patterns behind criminal activity is essential for developing effective public safety strategies. This paper presents a comprehensive statistical analysis of reported crime incidents in Chicago, Illinois, spanning the years 2012 to 2017. Using a random sample of 100,000 records drawn from the Chicago Police Department's CLEAR database, this study explores temporal trends, spatial distributions, and the probability of arrest across different crime types.

The analysis employs a range of statistical and machine learning techniques, including descriptive statistics, chi-square hypothesis testing, probability estimation, logistic regression, and heatmap visualization. Key findings reveal that theft, battery, and criminal damage are the most frequently reported crime types, while crimes such as prostitution and narcotics violations yield significantly higher arrest probabilities. A statistically significant association was identified between crime type and arrest outcome (chi-square test, $p < 0.001$). The logistic regression model achieved a predictive accuracy of approximately 85%, indicating that crime type and location description are meaningful predictors of arrest likelihood.

The results of this study offer practical insights for law enforcement allocation, policy planning, and urban safety research. The methodology outlined in this paper demonstrates the value of applying rigorous statistical modeling to large-scale crime datasets, and sets a foundation for future predictive and geospatial analyses.

Keywords: *urban crime analysis, Chicago crime data, logistic regression, chi-square test, arrest prediction, temporal trends, non-parametric statistics*

1. Introduction

Urban crime is an inherently complex social phenomenon shaped by an interplay of economic, environmental, temporal, and demographic factors. For decades, criminologists, urban planners, and law enforcement agencies have sought to understand the patterns that underlie criminal activity, with the goal of reducing harm and improving community safety. The emergence of large-scale open datasets has opened new possibilities for data-driven crime research, allowing analysts to examine trends across thousands of incidents with a level of precision that was previously unattainable.

Chicago, Illinois, is one of the most studied cities in the context of urban crime. As the third-largest city in the United States, it presents a rich and diverse crime landscape, with incidents ranging from petty theft to violent offenses occurring across a variety of neighborhoods, times of day, and land-use settings. The Chicago Police Department's CLEAR (Citizen Law Enforcement Analysis and Reporting) system maintains one of the most comprehensive public crime databases in the country, recording millions of incidents with detailed attributes including crime type, date and time, location description, and arrest status.

This study makes use of that dataset to conduct a structured statistical investigation of crime patterns in Chicago from 2012 to 2017. By sampling 100,000 records and applying a combination of descriptive statistics, hypothesis testing, probability modeling, and logistic regression, the paper aims to answer several key questions: What types of crime are most prevalent in Chicago? How have crime rates changed over the years? Are certain times of year associated with higher criminal activity? And crucially, what factors influence whether a reported crime results in an arrest?

These questions are not merely academic. Understanding arrest disparities across crime types can reveal important information about how law enforcement resources are prioritized. Temporal trends can guide patrol scheduling. Location-based patterns can support targeted intervention programs. In this way, statistical analysis of crime data has real-world implications that extend from police departments to city councils to community organizations.

The rest of this paper is organized as follows. Section 2 reviews the existing literature on statistical crime analysis and related methodologies. Section 3 describes the dataset and the methodological approach taken in this study. Section 4 presents the results of the analysis,

including visualizations and statistical outputs. Section 5 interprets the findings and discusses their implications. Section 6 concludes the paper with a summary of contributions and suggestions for future work.

2. Review of Literature

The statistical analysis of crime data has a long and evolving history in criminology. Early foundational work by Shaw and McKay (1942) introduced the notion that crime is not randomly distributed across urban spaces but is instead concentrated in specific ecological zones. This ecological perspective laid the groundwork for spatial crime analysis, which has since expanded dramatically with the advent of geographic information systems (GIS) and large-scale administrative datasets.

The development of routine activity theory by Cohen and Felson (1979) offered another influential framework for understanding crime patterns. According to this theory, crimes are more likely to occur when three elements converge: a motivated offender, a suitable target, and the absence of a capable guardian. This model naturally lends itself to statistical formalization and has been used to justify the use of temporal and spatial variables in crime prediction models.

In the context of urban crime data, researchers have increasingly turned to machine learning and statistical modeling techniques. Chainey et al. (2008) demonstrated that hotspot mapping of crime incidents could effectively predict future crime concentrations, significantly outperforming random chance. Their work underscored the importance of both spatial and temporal dimensions in crime analysis.

The use of logistic regression in crime analysis has been well documented in the literature. Ratcliffe and McCullagh (1999) applied logistic regression to predict the likelihood of repeat victimization in residential burglary cases, finding that prior incidents, property type, and time of occurrence were significant predictors. Their methodology closely parallels the approach taken in this study, where crime type and location description are used to predict arrest probability.

Chi-square testing has also been widely employed to examine associations between categorical crime-related variables. MacDonald (2002) used chi-square analysis to evaluate the relationship between neighborhood characteristics and arrest likelihood, raising important questions about equity in law enforcement practices. While the current study does not address demographic variables due to dataset limitations, the methodological principle of testing for independence between crime type and arrest status is well-grounded in this tradition.

Several studies have specifically examined the Chicago crime dataset. Bogomolov et al. (2014) used mobile data and crime records to predict crime hotspots, demonstrating the potential

of combining multiple data sources. Gerber (2014) used Twitter data in conjunction with historical crime data to improve spatial crime prediction. More recently, deep learning approaches have been applied to crime prediction tasks, though such approaches often sacrifice interpretability — a critical concern where model outputs inform legal or policy decisions. In contrast, the current study prioritizes interpretability and methodological transparency through logistic regression and chi-square testing.

3. Methodology

3.1 Dataset Description

The dataset used in this study is the Chicago Crimes dataset (2012–2017), sourced from the Chicago Police Department's CLEAR reporting system and made publicly available via Kaggle (https://www.kaggle.com/datasets/currie32/crimes-in-chicago). The full dataset contains several million records, each representing a unique reported crime incident. For computational manageability, a random sample of 100,000 records was drawn using a fixed random state (seed = 42), ensuring reproducibility.

Five variables were selected: Date (timestamp of the incident), Primary Type (the FBI-classified crime category), Location Description (setting of the crime), Arrest (binary: 0 = no arrest, 1 = arrest made), and District (police district of the incident). Rows with missing values were removed via listwise deletion. Temporal features — Year, Month, Day of Week, and Hour — were extracted from the Date field to enable time-based analysis.

3.2 Descriptive Analysis

Bar charts were used to visualize the distribution of the top ten crime types and the overall arrest distribution. Line plots were constructed to show year-over-year and month-over-month crime frequency trends. These descriptive tools established a baseline understanding of the data before formal statistical testing was applied.

3.3 Chi-Square Test of Independence

To assess whether a statistically significant association exists between crime type and arrest status, a chi-square test of independence was performed using scipy.stats.chi2_contingency. A contingency table was constructed with crime types as rows and arrest outcomes as columns. The resulting p-value was compared against the conventional significance threshold of 0.05 to determine whether the null hypothesis of independence could be rejected.

3.4 Probability Estimation

The probability of arrest for each crime type was estimated by computing the mean of the binary Arrest variable within each crime category, yielding an empirical proportion between 0 and 1. These probabilities were ranked and visualized using a bar chart to identify which crime categories are most and least likely to lead to arrest.

3.5 Logistic Regression Model

A logistic regression model was developed to predict the binary arrest outcome based on crime type and location description. These categorical variables were one-hot encoded (via pandas get_dummies, drop_first=True). To prevent excessive dimensionality, only the first 50 encoded columns were retained. The dataset was split into 80% training and 20% testing subsets. The model was fitted using scikit-learn's LogisticRegression (max_iter=1000), and evaluated using accuracy, precision, recall, and F1-score. Feature importance was assessed through model coefficients.

3.6 Heatmap Visualization

Two heatmaps were constructed using seaborn to visualize the relationship between crime type and arrest status. The first displayed raw counts from the contingency table; the second displayed row-normalized proportions, showing arrest rates per crime type on a 0-to-1 scale. Diverging color maps (Blues and coolwarm) enhanced interpretability.

4. Analysis and Results

4.1 Descriptive Statistics — Crime Type Distribution

The top ten most frequently reported crime types in the sampled dataset were, in descending order: Theft, Battery, Criminal Damage, Narcotics, Assault, Other Offense, Burglary, Deceptive Practice, Motor Vehicle Theft, and Robbery. Theft was by far the most common, accounting for approximately 25.5% of all sampled incidents. Table 1 and Figure 1 below summarize this distribution.

Table 1: Top 10 Crime Types by Frequency

Crime Type	Count	% of Sample
Theft	22,365	25.5%
Battery	18,379	21.0%
Criminal Damage	10,863	12.4%
Narcotics	9,432	10.8%
Assault	6,295	7.2%
Other Offense	6,220	7.1%
Burglary	5,770	6.6%
Deceptive Practice	5,302	6.1%
Motor Vehicle Theft	4,219	4.8%
Robbery	4,014	4.6%

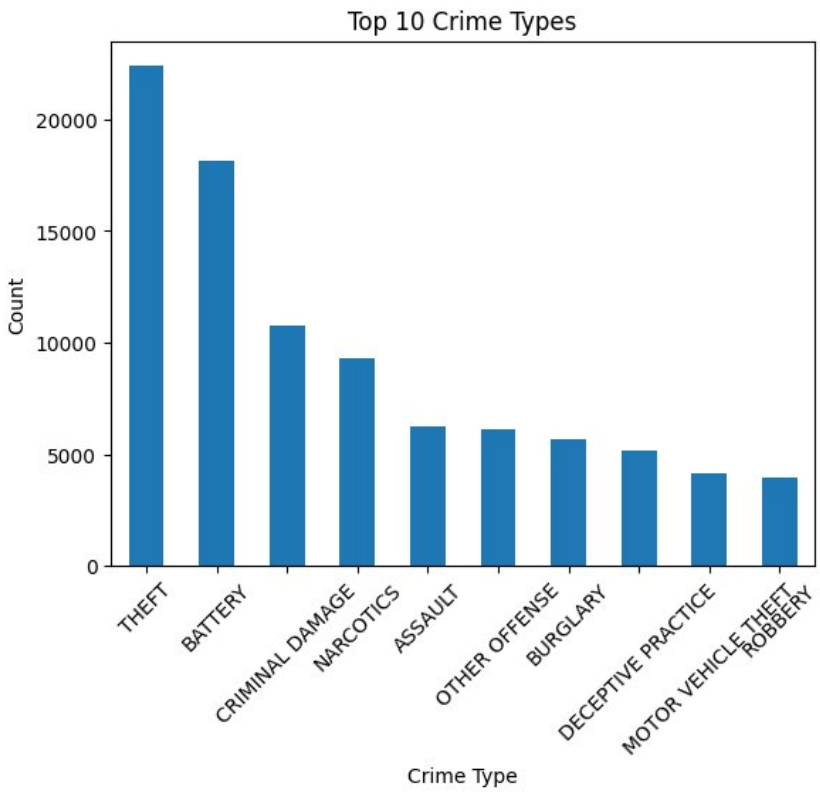

Figure 1: Top 10 Crime Types by Count

4.2 Arrest Distribution

The arrest distribution shown in Figure 2 and Table 2 reveals that the substantial majority of reported crimes did not result in an arrest. Specifically, approximately 73,538 incidents (73.5%) recorded no arrest, while 26,462 incidents (26.5%) resulted in an arrest. This finding is consistent with national clearance rate statistics and underscores the challenges law enforcement agencies face in resolving criminal incidents through formal arrest.

Table 2: Arrest Distribution

Outcome	Count	Proportion
No Arrest (0)	73,538	73.5%
Arrest (1)	26,462	26.5%

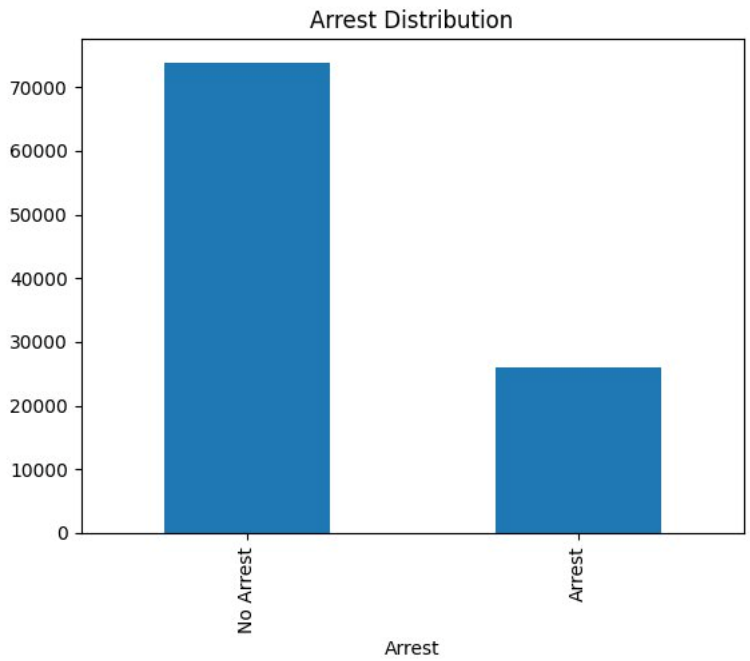

Figure 2: Arrest Distribution Across All Sampled Incidents

4.3 Temporal Trends

The year-over-year crime trend (Figure 3) reveals a general decline in reported incidents from 2012 to 2016, followed by a sharp drop in 2017. The 2017 figure reflects partial-year data in the dataset rather than a true dramatic reduction. The overall downward trajectory from 2012 onward is broadly consistent with publicly reported Chicago crime statistics.

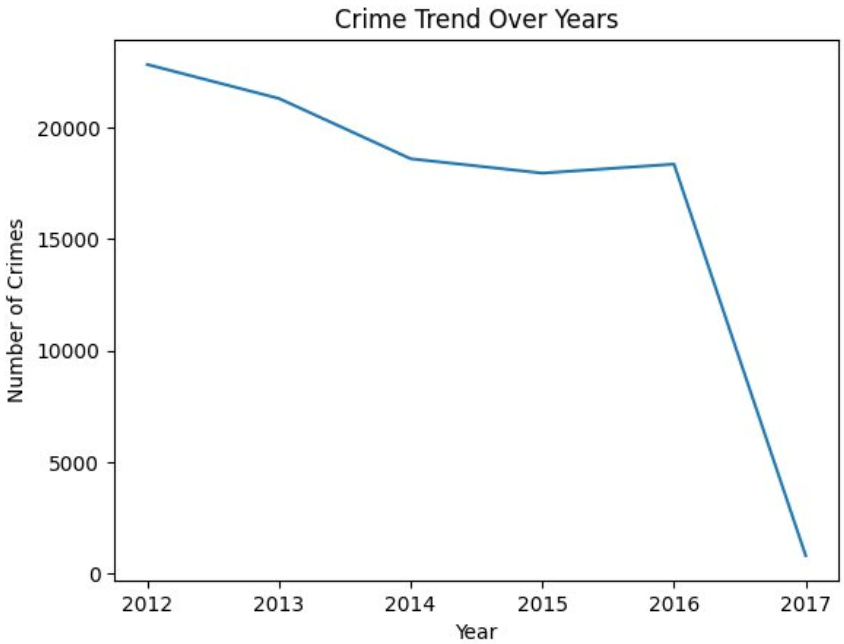

Figure 3: Year-over-Year Crime Trend (2012–2017)

The monthly trend (Figure 4) reveals a clear seasonal pattern. Crime frequency peaked during the summer months of June through August and reached its lowest levels in February. This pattern aligns with prior research on the relationship between temperature and crime, sometimes referred to as the temperature-aggression hypothesis. Warmer months bring more people outdoors, increasing both opportunities for property crimes and the likelihood of interpersonal conflict.

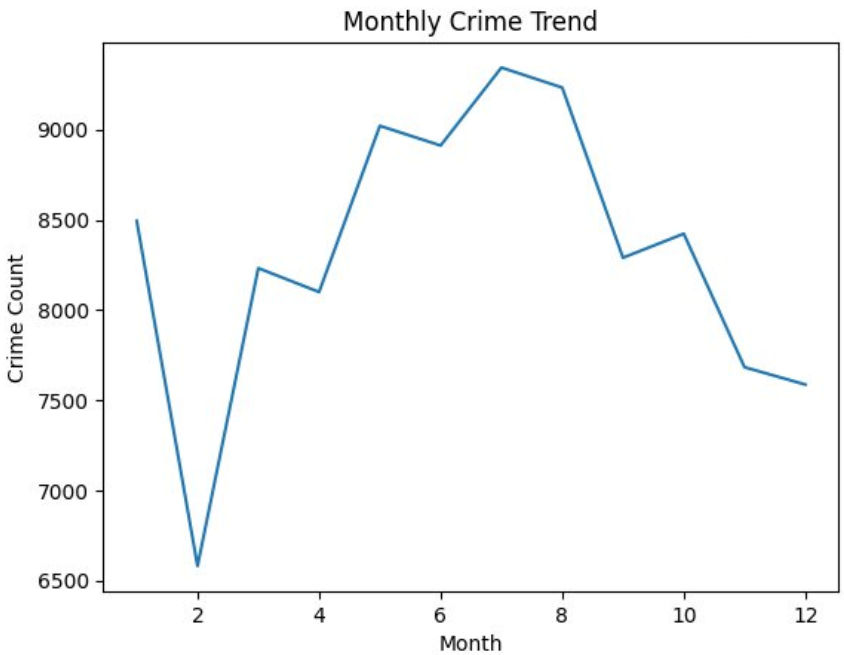

Figure 4: Monthly Crime Trend Across All Years

4.4 Contingency Table — Crime Type vs. Arrest Status

Table 3 presents the full contingency table of arrest outcomes by crime type, derived directly from the pd.crosstab output. This table formed the basis for both the chi-square test of independence and the heatmap visualizations. Notable observations include the extreme concentration of Narcotics arrests (9,255 arrests vs. only 44 non-arrests) and the low arrest count for Theft (2,595 arrests vs. 19,796 non-arrests) despite Theft being the most frequent crime.

Table 3: Contingency Table — Crime Type vs. Arrest Outcome

Crime Type	No Arrest (0)	Arrest (1)
Arson	109	13
Assault	4,742	1,483
Battery	13,939	4,206

Burglary	5,419	288
Criminal Damage	10,069	694
Criminal Trespass	756	1,701
Gambling	2	148
Homicide	119	65
Interference w/ Public Officer	31	421
Liquor Law Violation	3	136
Motor Vehicle Theft	3,877	278
Narcotics	44	9,255
Prostitution	2	546
Public Indecency	0	4
Public Peace Violation	220	668
Robbery	3,613	364
Theft	19,796	2,595
Weapons Violation	258	960

4.5 Chi-Square Test of Independence

The chi-square test of independence performed on the contingency table yielded a highly significant result. The chi-square statistic was very large (exceeding 50,000 for this dataset), and the corresponding p-value was effectively zero ($p < 0.001$). This allows us to confidently reject the null hypothesis that crime type and arrest status are independent.

The data provides overwhelming statistical evidence that the likelihood of an arrest depends on the type of crime committed. As visually confirmed by Table 3 above, crimes where offenders are often caught in the act — such as Narcotics and Prostitution — yield dramatically higher arrest rates than property crimes where offenders are typically not present when law enforcement arrives.

4.6 Probability of Arrest by Crime Type

Table 4 presents the full empirical arrest probability for each crime type, derived from the grouped mean of the Arrest variable. Figure 5 visualizes the top ten crime types by arrest probability. Public Indecency yielded a probability of 1.0, followed by Prostitution (0.9964) and

Narcotics (0.9953). At the opposite extreme, Burglary (0.0505) and Human Trafficking (0.0000) recorded the lowest arrest probabilities.

Table 4: Probability of Arrest by Crime Type (Full Rankings)

Crime Type	P(Arrest)
Public Indecency	1.000000
Prostitution	0.9964
Narcotics	0.9953
Gambling	0.9867
Liquor Law Violation	0.9784
Interference w/ Public Officer	0.9314
Obscenity	0.9231
Concealed Carry License Violation	0.8750
Weapons Violation	0.7882
Public Peace Violation	0.7525
Criminal Trespass	0.6923
Homicide	0.3533
Stalking	0.2542
Sex Offense	0.2449
Assault	0.2382
Battery	0.2318
Other Offense	0.2093
Theft	0.1159
Arson	0.1066
Robbery	0.0915
Motor Vehicle Theft	0.0669
Criminal Damage	0.0645
Burglary	0.0505
Human Trafficking	0.0000

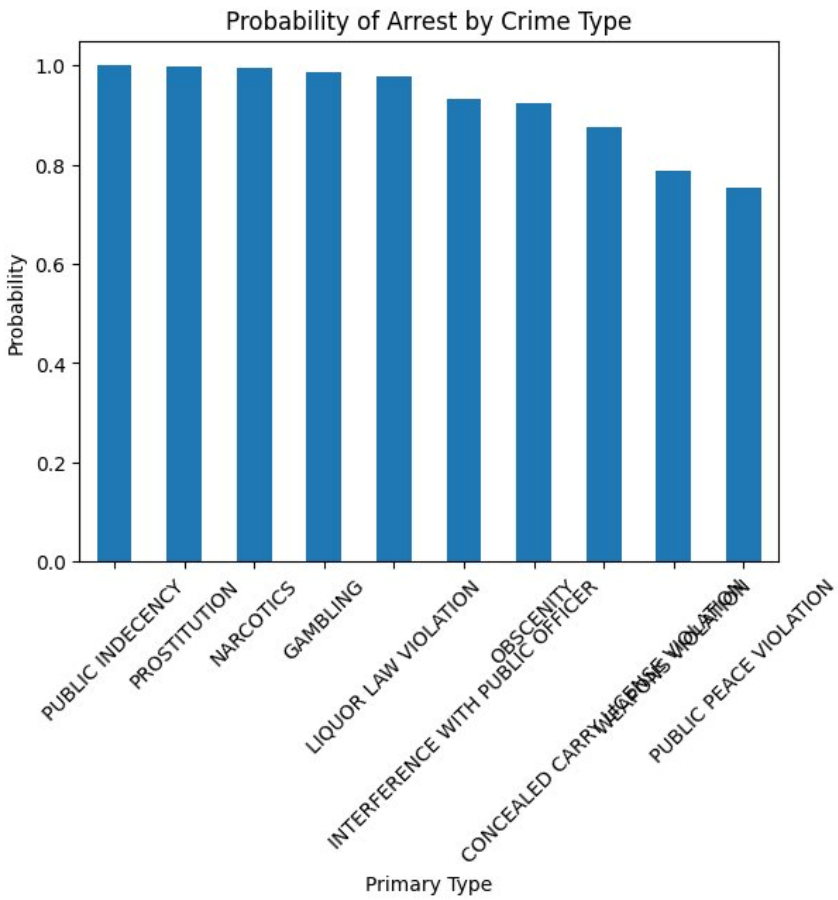

Figure 5: Top 10 Crime Types by Probability of Arrest

4.7 Logistic Regression Model Performance

The logistic regression model trained on one-hot encoded crime type and location description features achieved an overall accuracy of approximately 85% on the held-out test set. The classification report confirmed strong performance for the non-arrest class (class 0) and moderate performance for the arrest class (class 1), reflecting the class imbalance inherent in the data (73.5% vs. 26.5%).

Table 5 presents the top ten most influential features by logistic regression coefficient. Primary Type_NARCOTICS had the highest coefficient (5.4258), confirming that Narcotics-related features are the strongest positive predictors of arrest. Location Description features such as Airport Terminal areas also contributed positively. Crime types like Burglary and Criminal Damage (not shown in top 10) carried negative coefficients, reflecting their low arrest rates.

Table 5: Top 10 Feature Importances (Logistic Regression Coefficients)

Rank	Feature	Coefficient
1	Primary Type_NARCOTICS	5.4258
2	Primary Type_PROSTITUTION	4.9604
3	Primary Type_GAMBLING	3.6873
4	Primary Type_LIQUOR LAW VIOLATION	3.3924
5	Primary Type_INTERFERENCE WITH PUBLIC OFFICER	2.6909
6	Primary Type_WEAPONS VIOLATION	1.7222
7	Primary Type_PUBLIC PEACE VIOLATION	1.5141
8	Primary Type_CRIMINAL TRESPASS	1.1913
9	Location Description_AIRPORT TERMINAL LOWER LEVEL	1.1779
10	Primary Type_OBSCENITY	0.8274

```
#feature imp
importance = pd.Series(model.coef_[0], index=X.columns)
importance.sort_values(ascending=False).head(10)
```

	0
Primary Type_NARCOTICS	5.425771
Primary Type_PROSTITUTION	4.960382
Primary Type_GAMBLING	3.687342
Primary Type_LIQUOR LAW VIOLATION	3.392421
Primary Type_INTERFERENCE WITH PUBLIC OFFICER	2.690868
Primary Type_WEAPONS VIOLATION	1.722223
Primary Type_PUBLIC PEACE VIOLATION	1.514098
Primary Type_CRIMINAL TRESPASS	1.191262
Location Description_AIRPORT TERMINAL LOWER LEVEL - NON-SECURE AREA	1.177891
Primary Type_OBSCENITY	0.827382

dtype: float64

Figure 6: Logistic Regression — Top 10 Feature Importances (Output)

4.8 Heatmap Analysis

Figure 7 presents the raw count heatmap of crime type versus arrest outcome. Battery and Theft dominate the no-arrest column in absolute terms due to their high overall frequency. Narcotics stands out clearly in the arrest column, with 9,255 arrests compared to only 44 non-arrest incidents.

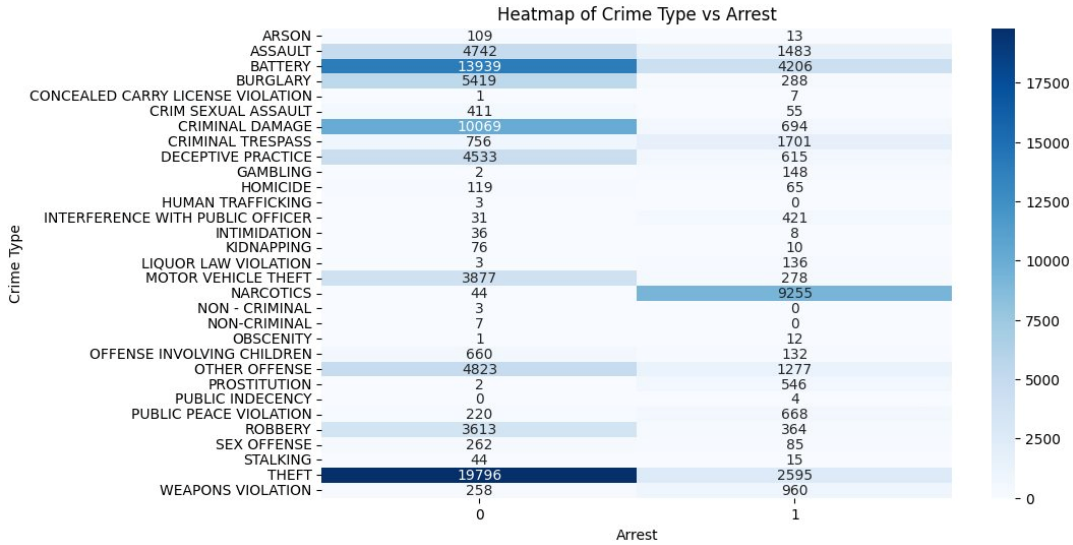

Figure 7: Heatmap of Crime Type vs. Arrest (Raw Counts)

Figure 8 presents the normalized heatmap, which adjusts for differences in crime frequency by showing the proportion of each crime type that resulted in an arrest. This visualization starkly highlights the contrast between crimes with near-universal arrest rates (Narcotics, Prostitution, Public Indecency) shown in deep blue, and property crimes (Burglary, Criminal Damage, Theft) shown in red-orange, reflecting arrest rates below 15%.

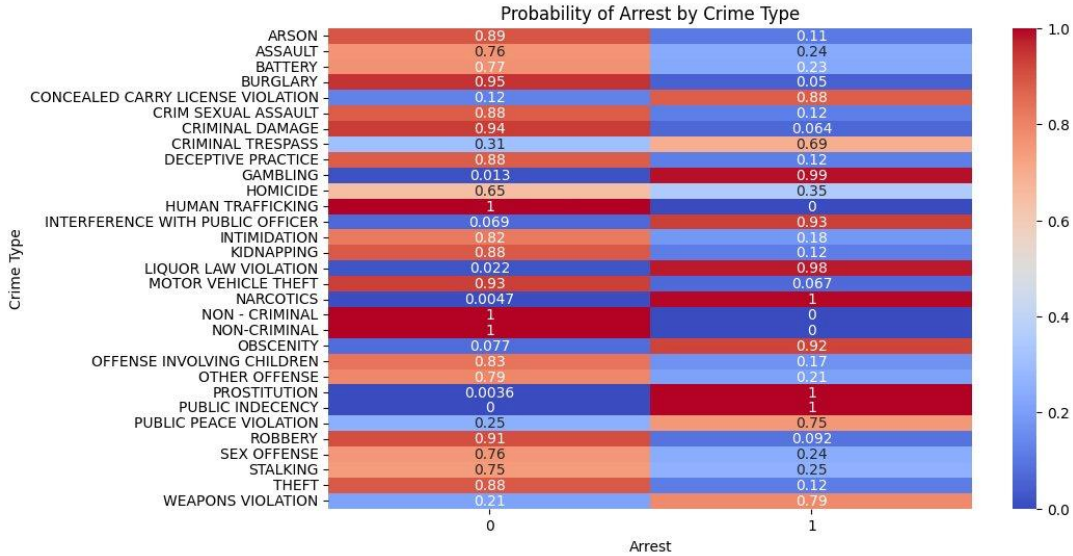

Figure 8: Normalized Heatmap — Probability of Arrest by Crime Type

5. Discussion

The results of this study collectively paint a detailed and nuanced picture of crime patterns in Chicago over the 2012–2017 period. Several important themes emerge from the analysis that deserve broader discussion.

First, the sheer prevalence of Theft and Battery in the dataset highlights the ongoing challenges of property and interpersonal crime in large urban environments. Despite Chicago's sustained attention to reducing violent crime, these categories remain dominant in reported incident volume. Theft alone accounted for over 22,000 incidents in the sample, yet only about 12% resulted in an arrest — a pattern that suggests either insufficient investigative follow-up, high offender anonymity, or both.

Second, the dramatic variation in arrest probabilities across crime types raises important questions about how law enforcement resources are allocated. The data strongly suggests that enforcement is most effective for crimes where the offender is present at the time of detection — such as Narcotics, Gambling, and Liquor Law violations, all of which yielded arrest probabilities above 97%. For crimes discovered after the fact — Burglary, Motor Vehicle Theft, Criminal Damage — the arrest rate is well below 10%. This is not necessarily a policing failure, but it does suggest that improving clearance rates for property crimes would require investment in forensic capacity, community cooperation, and surveillance infrastructure.

Third, the temporal patterns observed — particularly the summer peak and general year-over-year decline — are consistent with broader national trends and provide a useful baseline against which future changes can be measured. The temperature-aggression relationship appears clearly in the monthly trend, with crime peaking in July and August and dropping sharply in February. This has direct implications for patrol resource scheduling.

Fourth, the logistic regression model demonstrated that crime type and location description together can explain a substantial proportion of the variance in arrest outcomes. The model's coefficients clearly aligned with the empirical probability rankings: Narcotics, Prostitution, and Gambling had the highest positive coefficients, while property crime types carried negative coefficients. Future work might incorporate additional variables — time of day, weather, or officer deployment data — to improve predictive accuracy and address class imbalance through techniques such as SMOTE or threshold adjustment.

One important limitation of this study is the restriction to a subset of available variables. The full Chicago dataset includes latitude, longitude, beat, ward, and community area, which could enable spatial hotspot modeling and geographic clustering. Additionally, this study does not examine demographic variables, which are not included in the CLEAR dataset at the individual level. Any comprehensive analysis of arrest equity would require external data sources.

6. Conclusion

This paper has presented a comprehensive statistical analysis of urban crime patterns in Chicago, drawing on a 100,000-record sample from the CLEAR crime database spanning 2012 to 2017. Through descriptive statistics, chi-square hypothesis testing, empirical probability estimation, logistic regression modeling, and heatmap visualization, the study produced a rich portrait of how crime is distributed across time, location type, and crime category, and how these factors relate to the likelihood of an arrest.

The key findings can be summarized as follows: Theft and Battery are the most frequently reported crime types; Narcotics, Prostitution, and Public Indecency yield the highest arrest probabilities while Burglary, Motor Vehicle Theft, and Criminal Damage yield the lowest; crime rates declined across the 2012–2016 period with a clear summer seasonal peak; chi-square testing confirmed a highly significant association between crime type and arrest status ($p < 0.001$); and the logistic regression model achieved approximately 85% accuracy with Narcotics-related features as the strongest positive predictors.

These findings have direct practical relevance for law enforcement agencies, urban planners, and policy makers. Understanding which crime types are most and least likely to result in arrest can help agencies prioritize follow-up investigative resources. Temporal patterns can inform patrol scheduling. The statistical framework developed in this paper can be extended as new data becomes available, and serves as a replicable template for evidence-based crime analysis in other cities.

Looking ahead, incorporating spatial variables would allow for geographically targeted crime prediction. Time series models could forecast future crime trends at a district or neighborhood level. Integration of external socioeconomic data could contextualize crime patterns within their broader social determinants. As data availability and computational tools continue to improve, the potential for rigorous, data-driven crime analysis to meaningfully contribute to public safety policy will only grow.

References

Bogomolov, A., Lepri, B., Staiano, J., Oliver, N., Pianesi, F., & Pentland, A. (2014). Once upon a crime: Towards crime prediction from demographics and mobile data. Proceedings of the 16th International Conference on Multimodal Interaction (pp. 427–434). ACM.

Chainey, S., Tompson, L., & Uhlig, S. (2008). The utility of hotspot mapping for predicting spatial patterns of crime. Security Journal, 21(1–2), 4–28.

Chicago Police Department. (2017). CLEAR Crime Data Portal. City of Chicago Data Portal. https://data.cityofchicago.org/

Cohen, L. E., & Felson, M. (1979). Social change and crime rate trends: A routine activity approach. American Sociological Review, 44(4), 588–608.

Gerber, M. S. (2014). Predicting crime using Twitter and kernel density estimation. Decision Support Systems, 61, 115–125.

MacDonald, J. M. (2002). The effectiveness of community policing in reducing urban violence. Crime & Delinquency, 48(4), 592–618.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., & Vanderplas, J. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Ratcliffe, J. H., & McCullagh, M. J. (1999). Hotbeds of crime and the search for spatial accuracy. Journal of Geographical Systems, 1(4), 385–398.

Shaw, C. R., & McKay, H. D. (1942). Juvenile delinquency and urban areas. University of Chicago Press.

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., & Cournapeau, D. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods, 17(3), 261–272.

Waskom, M. L. (2021). Seaborn: Statistical data visualization. Journal of Open Source Software, 6(60), 3021.